Effect of warming on microbial decomposition of plant biomass in Arctic tundra soils using an stable isotope probing approach

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INTRODUCTION: Climate models predict that the Arctic will be significantly affected by climate change resulting in warmer summers and thicker winter snowpacks, both leading to higher soil temperatures. A warmer soil may promote plant growth, i.e. a higher amount of plant biomass (either litter or root exudates), and higher activity of microbial communities that play key roles in organic matter decomposition.

GOAL: The main aim of our study was to investigate the response of bacterial and fungal communities in the decomposition of plant-derived substrates in tundra soils under a scenario of increased summer temperatures and snow cover, using a stable isotope probing approach.

METHODS: In situ climate manipulation experiments were established in Western Greenland (Qeqertarsuaq Disko Island). Here we collected soil samples after eight years of warming treatment using open top chambers and snow fences. Collected soil samples were used for microcosm incubations with different ^{13}C-stable-isotopically labelled substrates (cellulose, glucose, acetate) to track the decomposition via respiration (analysis of $^{13}C\text{-}CO_2$) and incorporation into microbial biomass (analysis of ^{13}C-PLFAs and ^{13}C-fraction of DNA using 16S and ITS amplicon sequencing).

RESULTS: Results revealed that warming induced higher respiration of all substrates. Additionally, warming enhanced the incorporation of the substrates in bacterial biomass (PLFAs). Fungi highly incorporated the substrates, however there were less significant differences between warmed and control soils. To get a deeper understanding of the effect of warming on microbial communities, we complemented these results with DNA-SIP amplicon data.

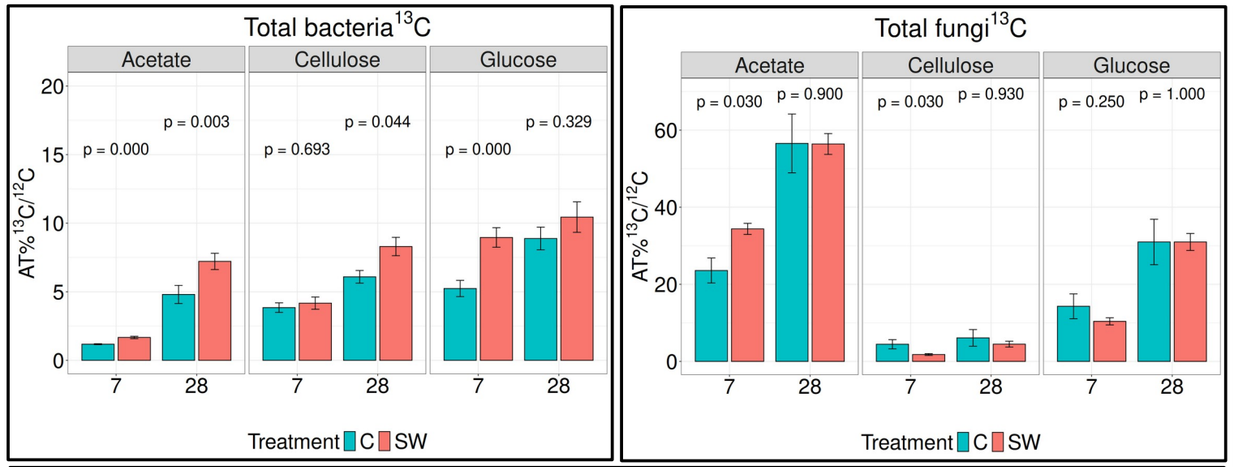

Figure 1. Incorporation of ^{13}C in the phospholipid fatty acids (PLFA) of microbial biomass. Data are shown for bacteria and fungi. The data represents the ratio between ^{13}C and ^{12}C that was incorporated in the biomass as PLFAs. Measurements were obtained via gas-chromatography mass-spectrometry (GC-MS).

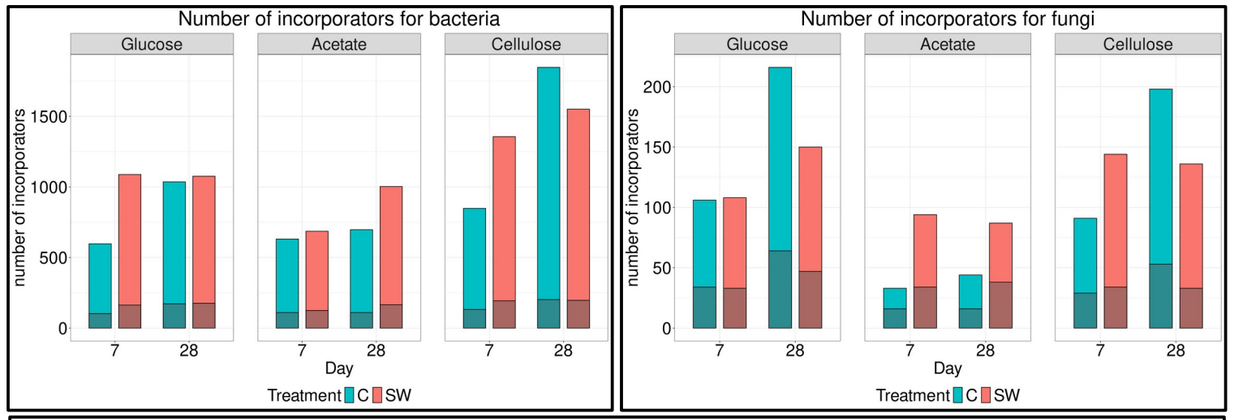

Figure 2. Number of ASVs (bacterial 16S) or clustered ASVs (fungal ITS), that incorporated one of the labelled substrates. Incorporators were retrieved using a quantitative SIP (qSIP) approach based on the HTSSIP R package. The shaded area at the bottom of each bar represents the number of unique Genera.

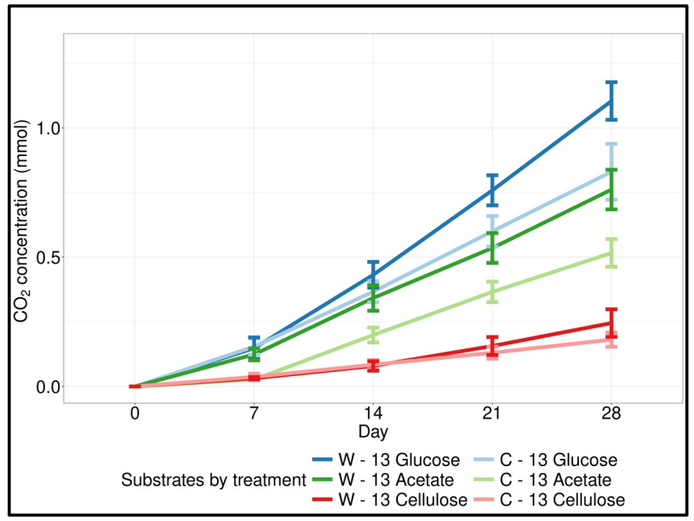

Figure 3. $^{13}C\text{-}CO_2$ production from the respiration of the labelled amended substrates: ^{13}C-glucose, ^{13}C-acetate, and ^{13}C-cellulose in treated (W) and control (C) soils.

CONCLUSION: This study contributes to the understanding of microbial decomposition of plant biomass stressing higher microbial respiration associated with warmed tundra soils of the Arctic.

The experimental site including snow fence simulating winter warming by increasing snow accumulation, open-top chambers simulating summer warming by increasing the temperature within the chambers, and the controls. Picture by Camelia Algora from Qeqertarsuaq (Disko Island, Greenland).

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